A UAV-based high-resolution vehicle trajectory dataset for motorcycle-dominated traffic

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Data Format v1.2 (last updated 02 June 2026)

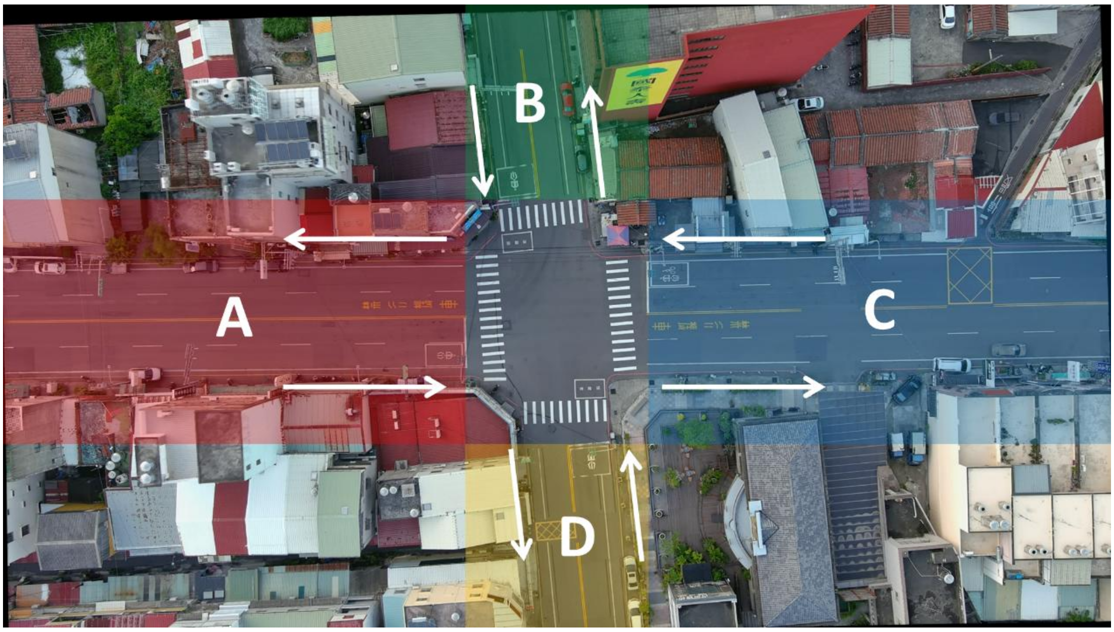
Basic descriptions

This file describes the files contained in the dataset. As of v1.2, the dataset contains trajectory data from 12 locations, and each location has 2 to 4 trajectory recordings associated with UAV video recordings.

For the location A01, there are following files:

- A metadata file (A01_metadata.docx)
- Background file for each video recordings (A01-01_background.jpg)
- Trajectory file (A01-01_TRJ.csv),
- Signal timing plan (A01-01_signal.csv)
- Line marking (A01-01_linemarking.csv; A01_linemarking.jpg)
- Video with traffic signal light information (upon request)

The four approaches of the intersection are named A, B, C and D. See the figure below.



1. Metadata files (_metadata.docx)

The metadata file contains the location and UAV-flight specific information of the data. Lane configuration, speed limit, and signal timing plan are also included.

1.1.Recording meta information

Name	**Description**	**Unit**
Location ID	The id of the recording location	[-]
Location Name	The correspond address of the recording location	[-]
Lat Coordinates	Rough latitude coordinates of recording location.	[deg]
Lon Coordinates	Rough longitude coordinates of recording location.	[deg]
Drone	The drone used for the recording	[-]
Flight altitude	The flight altitude of the recording	[m]
Frame Rate	The frame rate used for the recording	[fps]

1.2.Lane Configuration

Example:

Approach	**Lane 1**	**Lane 2**	**Lane 3**	**Lane 4**
A	LT	TR	-	-
B	LTR	-	-	-
C	LT	TR	-	-
D	LTR	-	-	-

L: left turn, LT, left turn and through, LTR, left turn, through, and right

1.3.Speed Limit

Example:

Approach A: 50 km/h

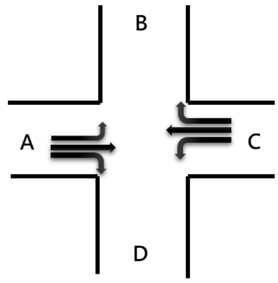
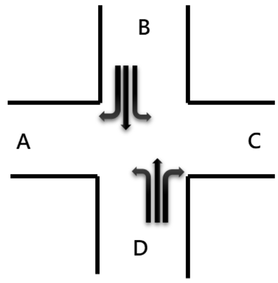
Approach B: 60 km/h

Approach C: 50 km/h

Approach D: 60 km/h

1.4.Signal Timing Plan

Example:

Cycle Length 80 s	Phase 1	Phase 2
Duration 16:00 ~ 17:30		
Green	26	40
Yellow	4	4
Red	3	3
Description	Direction A and Direction C Round Green	Direction B and Direction D Round Green

1.5.Each video recording information

Name	**Description**	**Unit**
Video	The id of the recording at the location	[-]
Record Time	The time the recording was done	[-]
Duration	The duration of the recording	[s]
Ground Sample Distance (GSD)	Scale factor from image pixels to meters.	[m/px]

2. Background (_background.jpg)

This file contains the clean background image (moving vehicles removed) of each UAV video data.

Example:

3. Trajectory files (_TRJ.csv)

This file contains the vehicle trajectory data of each video recordings.

Column	Description	Unit
Timestep	The time since the start in this record (10 hz)	[s]
Frame	The time frame since the start in this record	[-]
Vehicle ID	The id number of the vehicle in this record	[-]
Front X	X coordinate of the middle front bumper of the vehicle	[m]
Front Y	Y coordinate of the middle front bumper of the vehicle	[m]
Rear X	X coordinate of the middle rear bumper of the vehicle	[m]
Rear Y	Y coordinate of the middle rear bumper of the vehicle	[m]
Length	The length of the vehicle	[m]
Width	The width of the vehicle	[m]
Speed	The instantaneous forward speed of the vehicle	[m/s]
Acceleration	The instantaneous forward acceleration of the vehicle	$[m/s^2]$
intersection in	The direction of the vehicle approaching the intersection (AI, BI, CI, DI, X)	[-]
Intersection out	The direction of the vehicle exiting the intersection (AO, BO, CO, DO, X)	[-]
class	The class of the vehicle	[-]

Denotation of the class:

Letter	Type
b	Bus
c	Passenger car
g	Tractor-trailer (tractor)
h	Tractor-trailer (trailer)
m	Motorcycles (Powered-Two-Wheelers)
t	Truck
p	Pedestrian
u	Bicycle

The current release focuses on four vehicle classes: passenger cars, motorcycles, buses, and trucks. Other modes may be less reliable and can be excluded from analysis.

Example:

Timestep	Frame	Vehicle ID	Front X	Front Y	Rear X	Rear Y	Length	Width	Speed	Acceleration	intersection in	intersection out	class
0.2	1	0	113.118	13.777	108.094	14.091	4.653	2.215	0	3.9	X	X	c
0.2	1	1	38.975	34.344	34.344	34.501	4.649	1.854	0.55	-3.5	X	X	c
0.2	1	2	120.615	33.794	125.561	33.951	4.985	2.206	0.78	-3.4	CI	DO	c
0.2	1	3	85.8	84.388	85.486	80.776	3.557	2.251	0	0	X	X	c
0.2	1	4	124.266	63.428	124.58	58.796	4.615	2.358	0	0	X	X	t
0.2	1	5	58.522	50.79	54.597	50.632	4.194	1.981	0.39	-1.1	X	X	c
0.2	1	6	130.153	4.121	129.996	0.039	3.444	2.176	0.39	0	X	X	c
0.2	1	7	85.644	79.246	84.858	75.007	4.626	2.208	0.55	-2.7	X	X	c
0.2	1	8	75.988	64.37	75.752	59.189	5.106	2.24	7.46	2.1	X	X	c
0.2	1	9	74.536	47.924	73.986	43.214	4.704	2.157	8.27	-1.9	X	DO	c

4. Signal timing files (_signal.csv)

This file provides the signal timing data for each timestep corresponding to the trajectory data. Signal CSV values use r, y, and g for red, yellow, and green, and use '-' for movements that are geometrically impossible or not applicable at a location. Zero-duration signal rows are retained only when the metadata timing plan contains zero-duration phases. The offset between the start of the video and the signal timing plan has been accounted for through side-shot recording conducted on-site.

Column	Description	Unit
Timestep	Signal event start timestep	[s]
Duration	Signal event duration in seconds	[s]
AL	Direction A left turn signal phase	-
AT	Direction A through signal phase	-
AR	Direction A right signal phase	-
BL	Direction B left turn signal phase	-
BT	Direction B through signal phase	-
BR	Direction B right signal phase	-
CL	Direction C left turn signal phase	-
CT	Direction C through signal phase	-
CR	Direction C right signal phase	-

DL	Direction D left turn signal phase	-
DT	Direction D through signal phase	-
DR	Direction D right signal phase	-
A*	Protected Direction A motorcycle left-turn signal phase used in B06	-

Example:

Timestep	Duration	AL	AT	AR	Bl	BT	BR	CL	CT	CR	DL	DT	DR
0	16.26	g	r	r	r	r	r	r	r	r	g	r	r
16.26	4	y	r	r	r	r	r	r	r	r	y	r	r
20.26	3	r	r	r	r	r	r	r	r	r	r	r	r
23.26	26	r	r	r	g	r	r	g	r	r	r	r	r
49.26	4	r	r	r	y	r	r	y	r	r	r	r	r
53.26	3	r	r	r	r	r	r	r	r	r	r	r	r
56.26	40	g	r	r	r	r	r	r	r	r	g	r	r
96.26	4	y	r	r	r	r	r	r	r	r	y	r	r
100.26	3	r	r	r	r	r	r	r	r	r	r	r	r
103.26	26	r	r	r	g	r	r	g	r	r	r	r	r

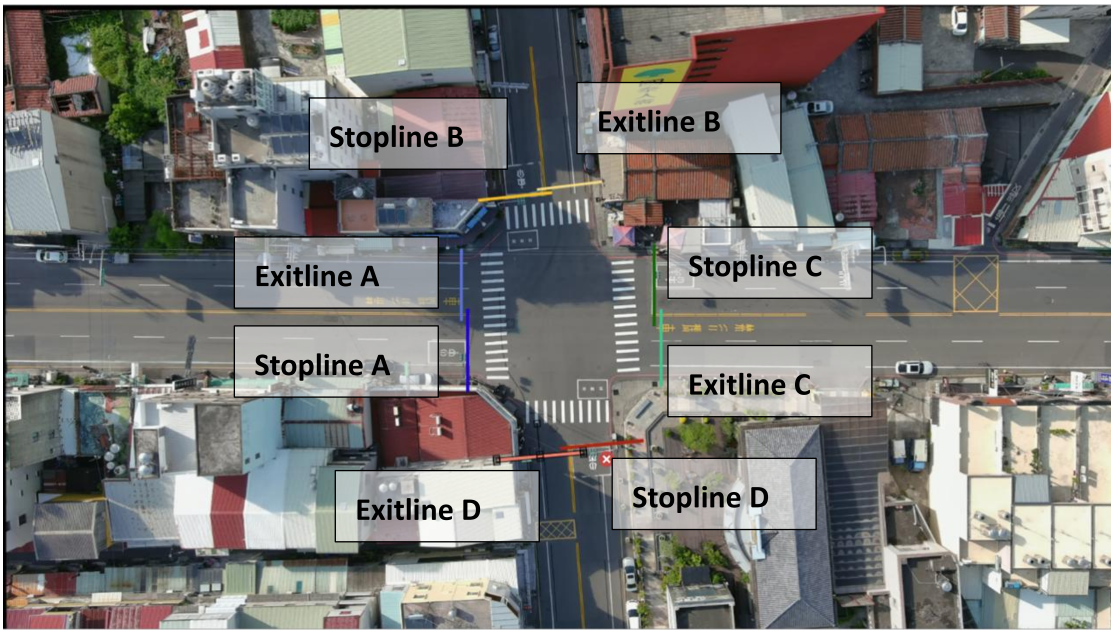
5. Line marking (_linemarking.csv; _linemarking.jpg)

This file contains the coordinates of the stop lines and exit lines for traffic entering and exiting the intersection. These coordinates are synchronized with the trajectory dataset. The accompanying JPG file illustrates the marked positions of these lines.

Example:

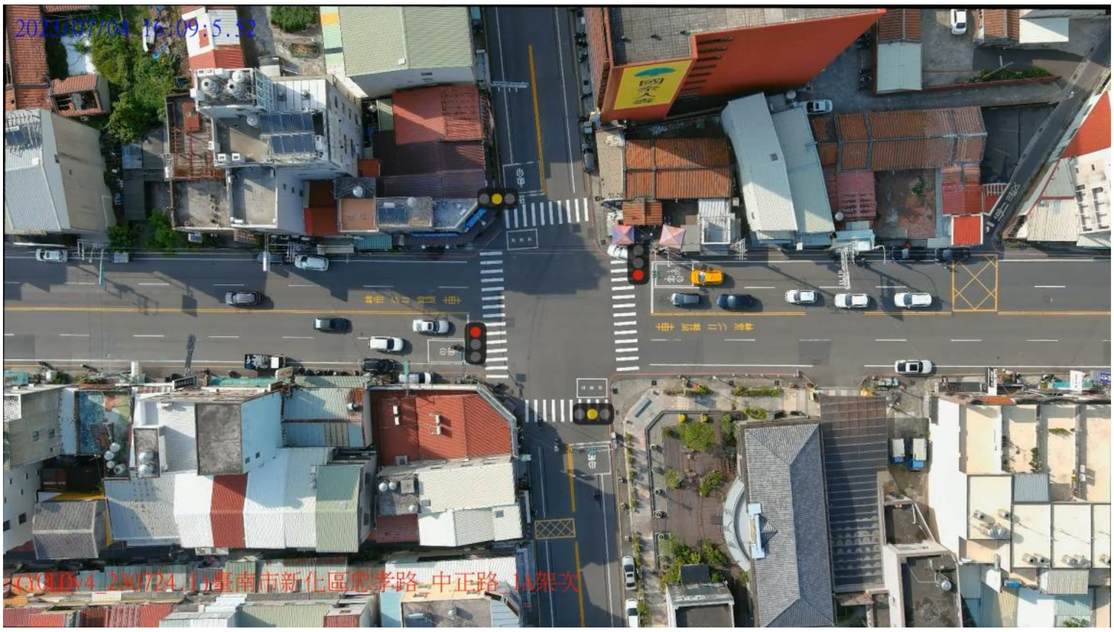
stopline				
Direction	X1	Y1	X2	Y2
A	62.75	41.66	63.35	63.7
B	63.63	26.36	73.89	25.26
C	88.45	31.19	88.1	42.38
D	76.95	59.93	86.97	58.91
exitline				
Direction	X1	Y1	X2	Y2
A	61.16	31.24	61.98	42.28
B	73.44	25.11	88.22	23.85
C	89.25	41.97	89.25	52.54
D	67.82	61.2	77.19	60.61

The marked positions of stop lines and exit lines for the four approaches:



6. Video with traffic signal light information (upon request)

The project also includes videos embedded with traffic signal light information. Due to their large file size, the videos are not included in the open dataset. Please contact the authors if you wish to access the videos.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2023/07/04 16:09:20.33
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